

8. Proof of Proposition 3

In the proof of Proposition 3, the lower case symbols q , r , and s represent elements that may be either a proof-parameter or a constant. By a “constant” in a proof, we mean any one of Section 5’s three possible representations of a constant, i.e. one of $\hat{c}[j]$ or either of an $\hat{u}[i]$ or $\hat{v}[i]$ appearing within the range of a $u\#[i,j]$ or $v\#[i,j]$ form.

Throughout this section, p denotes a proof of a Σ_1 sentence, denoted as Ψ . We will prove by induction that all p satisfy Proposition 3’s strong-uniform Σ_1 reflection principle. The proof’s overall structure will be similar to that of Proposition 1, although it will be a bit more elaborate.

For some Δ_0 formula $B(x_1, x_2 \cdots x_L)$, let us assume that (8.1) denotes Ψ .

$$\Psi = \{\exists x_1 \exists x_2 \cdots \exists x_L : B(x_1, x_2 \cdots x_L)\} \quad (8.1)$$

Define the Π_1 normalized contradiction of Ψ to be the Π_1 formula

$$\forall x_1 \forall x_2 \cdots \forall x_L : \neg B(x_1, x_2 \cdots x_L) \quad (8.2)$$

For any set of parameters or constants $s_1, s_2, \cdots s_L$, define the corresponding sharp contradiction of Ψ to be *any* formula of the form:

$$\neg B(s_1, s_2 \cdots s_L) \quad (8.3)$$

Define an intermediate contradiction of Ψ to be either its Π_1 -normalized contradiction or a formula of one of the $L-1$ forms

$$\begin{aligned} \forall x_2 \forall x_3 \forall x_4 \cdots \forall x_L : \neg B(s_1, x_2, x_3 \cdots x_L) \\ \forall x_3 \forall x_4 \cdots \forall x_L : \neg B(s_1, s_2, x_3 \cdots x_L) \\ \forall x_L : \neg B(s_1, s_2, \cdots s_{L-1}, x_L) \end{aligned} \quad (8.4)$$

The strong-uniform Σ_1 reflection principle for $IS^{\Sigma_1}(A)$ was defined in Section 3 to be the statement that all Σ_1 formulae Φ_v satisfy

$$\forall v \forall y \{ \text{Prf}_{IS^{\Sigma_1}(A)}(\ulcorner \Phi_v \urcorner, y) \supset \Phi_v^y \} \quad (8.5)$$

For any pair of parameters or constants r and q , define a primary and secondary deduction from the strong-uniform reflection axiom (8.5) to be a sentence similar to (8.6) and (8.7) respectively.

$$\forall y \{ \text{Prf}_{IS^{\Sigma_1}(A)}(\ulcorner \Phi_r \urcorner, y) \supset \Phi_r^y \} \quad (8.6)$$

$$\text{Prf}_{\text{IS}^1(A)}(\lceil \Phi_r \rceil, q) \supset \Phi_r^q \quad (8.7)$$

Our proof of Proposition 3 will employ some very close analogs of ANCESTOR(N) and INT from section 6. For our current discussion, ANCESTOR*(N) will be defined as the list of all the sentences in the proof p that lie in either the node N or one of its ancestors, *except that excluded are all the Group-3 axioms, their primary deductions, and all the intermediate contradictions of Ψ .* (Sharp contradictions of Ψ and secondary deductions from Group-3 axioms *are permitted* in ANCESTOR(N).) Once again, let us say N is **ancestor-consistent** iff there exists an "interpretation" function, denoted as INT^* , that assigns an integer value to each parameter symbol $\hat{u}[i]$ such that all $\text{IS}^1(A)$'s Group-1 and 2 axioms and all the sentences in ANCESTOR(N) are valid (under the Standard Model) when the symbols $\hat{u}[i]$ are assigned their designated integer values under INT^* . Again, N is called **maximal** if it is a node of maximal depth in the proof tree p possessing some such interpretation function.

Let N denote a maximal node and let N^* denote N 's child. In Section 6, N could have only one child, and N^* stored a sentence of the form $\neg \text{Prf}_{\text{IS}(A)}(\perp, q)$. Our present application is similar, and the carefully chosen definition of ANCESTOR*, in this case, forces N^* to store either a sharp contradiction of Ψ or a secondary deduction from a Group-3 axiom, (i.e. one of Equations (8.3) or (8.7)).

We are now ready to prove Proposition 3. Let p be an arbitrary proof of a Σ_1 theorem, and let N and N^* be as defined above. Since N^* must store a sentence of the form (8.3) or (8.7), we need merely to check that the *proof p satisfies the strong-uniform Σ_1 reflection property* for the two cases below:

Proof when p 's node N^* contains a sharp contradiction of Ψ : In this case $B(\text{INT}^*(s_1), \text{INT}^*(s_2), \dots, \text{INT}^*(s_L))$ must be true because otherwise $\neg B(s_1, s_2, \dots, s_L)$ would not provide the contradiction of INT^* (which by definition is associated with N^*). Moreover since Addition is the only increasing function available in the proof p , it must be true that each $\text{INT}^*(s_i) < p$ (for the same reasons that the similar bound appeared in the proof of Proposition 1). Hence, each of the s_i are small enough for the strong-uniform Σ_1 reflection principle to be valid for the proof p . QED

Proof for the "ALTERNATE" case where p 's node N^* contains a secondary deduction of a Group-3 axiom: To prove strong-uniform Σ_1 reflection for this case, it is sufficient to show that it is impossible for this case to occur.

Our proof will begin by inductively assuming that all Gödel numbers $i < p$ satisfy strong-uniform Σ_1 reflection. If the sentence appearing in N^* was Equation (8.7)'s secondary deduction, then $\text{Prf}_{\text{IS}^{\Sigma_1}(A)}(\ulcorner \Phi_{\text{INT}^*(q)} \urcorner, \text{INT}^*(q)) \supset \Phi_{\text{INT}^*(q)}$ would be *false* (because the interpretation INT^* contradicts the sentence $\text{Prf}_{\text{IS}^{\Sigma_1}(A)}(\ulcorner \Phi_i \urcorner, q) \supset \Phi_i^q$ in this case). Once again, the slow growth rate of the Addition function implies that $\text{INT}^*(q) < p$. Thus, some integer $i < p$ must violate the strong-uniform Σ_1 reflection principle. The preceding statement contradicts the inductive hypothesis, and hence shows this “alternate” case can not occur. QED

Remark 3. Define a $G\Sigma_1$ formula to be a wff $\Omega = \{ Q_1 x_1 Q_2 x_2 \cdots Q_j x_j : \Theta(x_1 x_2 \cdots x_j) \}$ where each Q_i is either an existential quantifier or a bounded universal quantifier, and where Θ is Δ_0 formula. Proposition 3 can be strengthened to indicate the consistency of a system $\text{IS}^{G\Sigma_1}(A)$, whose Group-3 axioms indicate that the strong uniform Σ_1 reflection principle holds for all $G\Sigma_1$ formulae (see footnote *)

Remark 4. Propositions 1 & 3 hold both when the language of $\text{IS}^{\Sigma_1}(A)$ uses an m -ary function symbol $f(x_1, x_2, \dots, x_m)$ to denote a function, and when it is represented by an $(m+1)$ -ary relation symbol $A_f(x_1, x_2, \dots, x_m, w)$ together with the added axiom $\forall x_1 \forall x_2 \dots \forall x_m \exists w : A_f(x_1 x_2 \dots x_m w)$. (Only for the sake of simplifying the proof notation, Section 6 had assumed the latter during its growth analysis in Equations (6.2) thru (6.4).)

9. Proof of Proposition 4 and a Discussion of its Generalizations

The proof of Proposition 4 is quite simple. Lemma 5.3 implies that $\text{IS}^{\Sigma_1}(\text{PA}+)$ can prove (9.1) for any fixed i .

$$\{\exists x \text{Prf}_{\text{IS}^{\Sigma_1}(\text{PA}+)}(\ulcorner \forall v \Phi(v) \urcorner, x)\} \supset \{\exists y \text{Prf}_{\text{IS}^{\Sigma_1}(\text{PA}+)}(\ulcorner \Phi_i \urcorner, y)\} \quad (9.1)$$

Also $\text{IS}^{\Sigma_1}(\text{PA}+)$'s Σ_1 reflection property implies that it can verify (9.2) when Φ_i is Σ_1 .

$$\{\exists y \text{Prf}_{\text{IS}^{\Sigma_1}(\text{PA}+)}(\ulcorner \Phi_i \urcorner, y)\} \supset \Phi_i \quad (9.2)$$

Since $\text{IS}^{\Sigma_1}(\text{PA}+)$ can prove (9.1) & (9.2) (for any Φ_i which is Σ_1), the Cut Elimination Theorem implies $\text{IS}^{\Sigma_1}(\text{PA}+)$ can also prove (9.3).

$$\{\exists x \text{Prf}_{\text{IS}^{\Sigma_1}(\text{PA}+)}(\ulcorner \forall v \Phi(v) \urcorner, x)\} \supset \Phi_i \quad \text{QED} \quad (9.3)$$

* The formal proof of the footnoted statement requires some tedious notation, but it is easy to explain the intuition behind it. It intuitively follows from the fact that every $G\Sigma_1$ formula Ω can be mapped onto a Σ_1 formula Ω^* such that $\Omega \equiv \Omega^*$ under the natural interpretation of bounded quantifiers.

The perhaps most interesting facet of Proposition 4 and the proof above is that they can be generalized significantly. Let Ψ denote an arbitrary Π_{2n} sentence of the canonical form (9.4), where Θ below is a Δ_0 formula.

$$\Psi = \{ \forall x_1 \exists y_1 \forall x_2 \exists y_2 \dots \forall x_n \exists y_n : \Theta(x_1 x_2 \dots x_n y_1 y_2 \dots y_n) \} \quad (9.4)$$

Let $I = (c_1, c_2 \dots c_n)$ denote an arbitrary n -tuple of integer constants. Define $\text{Branch}_\Psi(p, I)$ to be the Σ_1 formula $\{ \exists y_1 \exists y_2 \dots \exists y_n : \Theta(c_1 c_2 \dots c_n y_1 y_2 \dots y_n) \}$ with the added caveat that each y_i is required to satisfy the constraint $y_i < p \cdot \text{Max}(c_1, c_2 \dots c_i)$. Given a set of m distinct n -tuples $I_1 I_2 \dots I_m$, define its E-Tree, denoted as $\text{Tree}_\Psi(p, I_1, I_2 \dots I_m)$, to be the Δ_0 formula which asserts that the conjunction of the formulae $\text{Branch}_\Psi(p, I_1), \text{Branch}_\Psi(p, I_2) \dots \text{Branch}_\Psi(p, I_m)$ is true, *and additionally* if each of the first k constants in I_i and I_j are identical then the first k of the $y_1 y_2 \dots y_k$ values in $\text{Branch}_\Psi(p, I_i)$ and $\text{Branch}_\Psi(p, I_j)$ are required to be matching.

An axiom system B 's **E-Tree Reflection Principle** for Ψ and $(I_1, I_2 \dots I_m)$ is the sentence:

$$\forall p \{ \text{Prf}_B(\ulcorner \Psi \urcorner, p) \supset \text{Tree}_\Psi(p, I_1, I_2 \dots I_m) \} \quad (9.5)$$

Also, its **Bounded E-Tree Reflection Principle** for Ψ , $(I_1, I_2 \dots I_m)$ and a fixed exponent λ is the assertion:

$$\forall p \{ [(\exists x = p^\lambda) \wedge \text{Prf}_B(\ulcorner \Psi \urcorner, p)] \supset \text{Tree}_\Psi(\infty, I_1, I_2 \dots I_m) \} \quad (9.6)$$

In [Wi94], we generalize the techniques of Sections 7 & 8 to prove:

Proposition 8. For each nice system A , there exists an introspective system $\text{IS}^{\text{ETR}}(A)$, which is consistent and whose Group-3 axioms are strong enough to verify its E-Tree Reflection Principle for *every* Ψ and $(I_1, I_2 \dots I_m)$.

The proof of Proposition 8 has been postponed until [Wi94] because of its length. However, it is possible to sketch a short proof for Proposition 9 below (which is only slightly weaker than Proposition 8).

Proposition 9. Let A denote either the axiom system PA^+ , or any extension of it. Then the system $\text{IS}^{\Sigma_1}(A)$ can prove the validity of its Bounded E-Tree Reflection Principle for *every* Ψ and $(I_1, I_2 \dots I_m)$ with $\lambda = m$ in Equation (9.6).

Very Informal Proof Sketch. Employing an analog of Lemma 5.3's proof, the system A can associate a *fixed* constant K with each *fixed* tuple $(I_1, I_2 \dots I_m)$, such that if p is a proof of Ψ then there exists $q < K \cdot p^m$ which is a proof of $\text{Tree}_\Psi(\infty, I_1, I_2 \dots I_m)$. As before, the preceding remark can be encoded as a Π_1 sentence, implying that the Group-2 axioms will cause it to be a theorem

of $IS^{\Sigma_1}(A)$. Since $IS^{\Sigma_1}(A)$ recognizes scalar multiplication as a function, it can thus prove:

$$\forall p \forall x \{ [x = p^m \wedge \text{Prf}_{IS^{\Sigma_1}(A)}(\ulcorner \Psi \urcorner, p)] \supset \exists q \text{Prf}_{IS^{\Sigma_1}(A)}(\ulcorner \text{Tree}_{\Psi}(\infty, I_1, I_2 \dots I_m) \urcorner, q) \} \quad (9.7)$$

Moreover, $IS^{\Sigma_1}(A)$'s strong-uniform Σ_1 reflection axioms indicate that

$$\exists q \text{Prf}_{IS^{\Sigma_1}(A)}(\ulcorner \text{Tree}_{\Psi}(\infty, I_1, I_2 \dots I_m) \urcorner, q) \supset \text{Tree}_{\Psi}(\infty, I_1, I_2 \dots I_m) \quad (9.8)$$

Hence, $IS^{\Sigma_1}(A)$ can prove (9.6) with $\lambda = m$, by the Cut Elimination Theorem and the fact that $IS^{\Sigma_1}(A)$ can prove (9.7) & (9.8). (A more detailed justification of Propositions 8 & 9 appears in [Wi94].)

10. Philosophical Implications of Propositions 2c, 4, 8 and 9

The significance of Propositions 2c, 4, 8 and 9 is perhaps not quite self-evident. Clearly these reflection principles provide some form of information not available in classical logics, since Lob's Theorem [Me87] states that traditional axiom systems are unable to prove their reflection principles, except in the trivial case where the right side of the implicant is also provable. On the other hand, since every Δ_0 sentence is decidable from merely the Group 1 axioms and since all valid Σ_1 sentences are provable from Group 1, these reflection principles have only limited implications in extending *what one is capable of learning*. Their main significance lies in unifying the separate proofs for each $\Phi_1, \Phi_2, \Phi_3 \dots$. For instance by Proposition 4, a single proof for the Π_2 sentence $\forall v \Phi(v)$ is *known INTERNALLY* by introspective systems *to have a reflection power* comparable to more expensive separate proofs for each of its Σ_1 instances $\Phi_1, \Phi_2, \Phi_3 \dots$. (The E-trees in Proposition 8 & 9 generalize this effect.)

The most surprising facet of Propositions 2c, 4, 8 and 9 will be their stark contrast with Proposition 5 (whose proof appears in the next section). Proposition 5 asserts that the local reflection sentences for certain Π_1 sentences $\forall v \Phi(v)$ can not be presumed by an IS-like system, while Propositions 2c, 4, 8 & 9 show that the analogous "*semi-uniform*" reflection principle for its Δ_0 instance set $\Phi_1, \Phi_2, \Phi_3 \dots$ does *nevertheless* (!!!) automatically follow.

One way to appreciate philosophically Propositions 2c, 4, 8 and 9 is to view man as living a quite mundane bounded life, where he is technically able to recognize and perceive the truth of *only very tangible* Δ_0 sentences. In a sense (formalized by Proposition 5), even the simplest form of infinite sentences (of the Π_1 type) are beyond the human's horizon for comprehension. However, *the proof* of a Π_n theorem, unlike *its formal statement*, appears to be technically (!!!) within man's intellectual grasp because it helps to more orderly organize the world of Δ_0 knowledge (in a sense formalized by Propositions 2c, 4, 8 and 9).

11. Proof of Proposition 5

In this section, the keyword INVALID.1 will designate the axiom system $IS^{L\Pi_1}(A)$, where A is either $PA+$ or an extension of it. (A second variant of “invalid axioms system” will be discussed in Section 12.) This section will prove INVALID.1 is inconsistent by applying Godel’s paradigm to essentially the sentence asserting “*Every proof x from the axiom system INVALID.1 fails to verify this sentence*”. The combination of the fact that this sentence can be encoded as a Π_1 statement and the Π_1 Local Reflection Principle will be used to prove INVALID.1 must be inconsistent.

Define $SUBST(a,b)$ to be the classic Δ_0 formula expressing that if a is the Godel number of $\Phi(x)$ then $SUBST(a,b)$ is satisfied only by that b which is the Godel number of the sentence obtained by replacing x with the constant a (see *). The italicized sentence from the previous paragraph can be formally represented by taking Φ from (11.1) and forming the sentence $\Phi(\ulcorner\Phi\urcorner)$.

$$\Phi \equiv \{ \forall x_2 \forall x_3 [SUBST(x_1, x_2) \supset \neg \text{Prf}_{\text{INVALID.1}}(x_2, x_3)] \} \quad (11.1)$$

Let H denote the special integer satisfying

$$SUBST(\ulcorner\Phi\urcorner, H). \quad (11.2)$$

INVALID.1 can obviously prove that $\Phi(\ulcorner\Phi\urcorner)$ is equivalent to (11.3).

$$\Psi \equiv \{ \forall x \neg \text{Prf}_{\text{INVALID.1}}(H, x) \} \quad (11.3)$$

A semantic tableaux proof of Ψ appears in Figure 3. Let $\bar{q}$ denote this proof, and $\bar{r}$ denote INVALID.1’s proof that Ψ is equivalent to $\Phi(\ulcorner\Phi\urcorner)$. Since INVALID.1 can recognize both $\bar{q}$ and $\bar{r}$ as *fixed constants*, it must be able to recognize a third Godel number $\bar{p}$, corresponding to a proof of $\Phi(\ulcorner\Phi\urcorner)$. (To be precise, the existence of $\bar{p}$ is an immediate consequence of the existence of $\bar{q}$ and $\bar{r}$ by the Meta-Logic, described in Remark 2 of Section 7). **THE MAIN POINT IS** that $\Phi(\ulcorner\Phi\urcorner)$ asserts it has no such proof $\bar{p}$! This implies INVALID.1 is inconsistent. Or if one wishes to be strictly formal, the precise paradox is that:

Since $\bar{p}$ is a *fixed constant*, representing a proof of $\Phi(\ulcorner\Phi\urcorner)$, there is a *formal proof* in INVALID.1 of the *theorem statement* that $\bar{p}$ is a *proof* of $\Phi(\ulcorner\Phi\urcorner)$. This latter theorem formally contradicts $\Phi(\ulcorner\Phi\urcorner)$, which is another *theorem of INVALID.1* ! Hence, INVALID.1 is inconsistent.

* While essentially all interesting introspective axiom systems are too weak to prove b is functionally determined by a under the $SUBST(a,b)$ relation, they nevertheless have no difficulty in expressing $SUBST(a,b)$ as a Δ_0 formula. The formal encoding of $SUBST$ follows from the combination of Lemma 5.1. and Theorem A.1 (from Appendix A).

12. Justification of Proposition 6.

Let the keyword INVALID.2 be an abbreviation for Proposition 6's axiom system $IS^{\cup \Delta_0}(PA+, G_m)$. The following lemma will help us prove INVALID.2's inconsistency.

Lemma 12.1. All of INVALID.2's local reflection statements for Π_1 sentences are theorems of this axiom system.

Proof Sketch: For each Δ_0 formula $\Psi(v)$ with only v free, there exists some constant $\lambda > 0$ such that PA+ can prove (12.1) as a theorem. (One needs to merely apply Lemma 5.3's T-transformation to verify this fact.)

$$\forall a \forall b \forall c \exists d < c : \{c = (ab)^\lambda \wedge \text{Prf}_{\text{INVALID.2}}(\ulcorner \forall v \Psi(v) \urcorner, a)\} \supset \text{Prf}_{\text{INVALID.2}}(\ulcorner \Psi_b \urcorner, d) \quad (12.1)$$

Since (12.1) is a Π_1 theorem of PA+, the class of Group-2 axioms imply that it will also be a theorem of INVALID.2. Since INVALID.2 recognizes multiplication as a total function, another theorem of it will clearly be $\{\forall a \forall b \exists c c = (ab)^\lambda\}$. These two facts (via another application of Meta-Logic) imply that (12.2) must also be a formal theorem of INVALID.2.

$$\forall a \forall b \exists d : \text{Prf}_{\text{INVALID.2}}(\ulcorner \forall v \Psi(v) \urcorner, a) \supset \text{Prf}_{\text{INVALID.2}}(\ulcorner \Psi_b \urcorner, d) \quad (12.2)$$

Moreover, the proof of Ψ_b is known *within* INVALID.2 to imply Ψ_b 's validity (in the "uniform" sense), by INVALID.2's $U \Delta_0$ reflection property. Thus, it can verify

$$\forall b \forall d : \text{Prf}_{\text{INVALID.2}}(\ulcorner \Psi_b \urcorner, d) \supset \Psi(b) \quad (12.3)$$

Since INVALID.2 can prove (12.2) & (12.3), the Cut Elimination Theorem, via Remark 2's perspective (from page 22), implies it can also prove (12.4):

$$\forall a : \text{Prf}_{\text{INVALID.2}}(\ulcorner \forall v \Psi(v) \urcorner, a) \supset \forall v \Psi(v). \quad (12.4)$$

Since Ψ is an arbitrary Δ_0 formula, the preceding argument shows that INVALID.2 can prove all its local Π_1 reflection statements. QED

The main part of the proof of Proposition 6 is nearly the same as Proposition 5's proof. This is because Proposition 5 implied that any introspective system which includes the $L\Pi_1$ reflection axioms is automatically inconsistent, and Lemma 12.1 indicated that the $L\Pi_1$ reflection statements are indeed theorems of INVALID.2.

The previous paragraph contained the phrase "*nearly the same proof*" because one remaining difference between Propositions 5 & 6 is that INVALID.1 treats $L\Pi_1$ reflection as an *axiom schema*, whereas Lemma 12.1 only established the $L\Pi_1$ reflection sentences as *theorems of* INVALID.2.

This fact implies that the *precise* semantic tableaux in Figure 3 is not helpful for proving Proposition 6 because the third line in Figure 3's tableaux is technically permissible *only when* $L\Pi_1$ reflection is an axiom. This detail is mostly a technical rather than conceptual distinction between Propositions 5 & 6: Appendix B explains formally how after one minor modification the exact formalism from Chapter 11 can be extended to also prove Proposition 6.

13. Proof of Proposition 7 and Added Remarks.

Define a tableaux proof with cuts to be a deduction formalism identical to semantic tableaux, except that it is permissible to include the sentence $\Phi \vee \neg\Phi$, for any sentence Φ , in any node of a proof tree. Although this section proves Proposition 7 formally for tableaux with cuts, its proof trivially generalizes to all the other usual deduction methods allowing cuts.

In our proof, Ψ will denote $IS_+(A, G_-)$'s Group-3 axiom sentence, indicated below.

$$\Psi \equiv \{ \forall x \neg \text{Prf}_{IS_+(A, G_-)}(\perp, x) \} \quad (13.1)$$

Let $\text{Model}(k)$ denote the finite model generated by the first $k+1$ natural numbers $0, 1, 2, \dots, k$ under the functions and relations of the axiom group G_- . (This definition is reasonable because all the functions f of G_- satisfy $f(x_1, x_2, \dots, x_n) \leq \text{Max}(x_1, x_2, \dots, x_n)$.) Let G_-^k be a set of Group-1 axioms identical to G_- , except that G_-^k discusses only the first $k+1$ constants $\bar{0}, \bar{1}, \bar{2}, \dots, \bar{k}$. Also let $\text{Schema}(k)$ denote the union of G_-^k with $IS_+(A, G_-)$'s Group-2 axioms. Since $\text{Schema}(k)$ is valid under $\text{Model}(k)$, $IS_+(A, G_-)$ can be inconsistent only if there exists some integer k such that Ψ is invalid under $\text{Model}(k)$. We will prove that this can not occur by assuming, for the sake of contradiction, that n is the minimal integer where $\neg\Psi$ is valid under $\text{Model}(n)$.

This assumption implies that there exists some $t \leq n$ in $\text{Model}(n)$ such that t satisfies

$$\text{Prf}_{IS_+(A, G_-)}(\perp, t) \quad (13.2)$$

Let m denote the largest constant appearing in the proof t , and let $\text{Schema}(m)$ and $\text{Model}(m)$ have the same definitions as $\text{Schema}(k)$ and $\text{Model}(k)$. By definition, Ψ is the only possible axiom which may appear in the proof t and which is not in $\text{Schema}(m)$. Since all the axioms of $\text{Schema}(m)$ are valid under $\text{Model}(m)$, the combination of (13.2) and the previous sentence imply that Ψ is invalid under $\text{Model}(m)$. Clearly m is less than t , since it appears in the proof t . The last two sentences provide the desired contradiction because $m < t \leq n$, and n was presumed to be the least integer such that $\neg\Psi$ is valid under $\text{Model}(n)$. QED

Remark 5. Some hybrid versions of Proposition 7 that follow from the techniques of Sections 8-11 are that $IS_+^{\Sigma_1}(A, G_-)$, $IS_+^{U\Sigma_1}(A, G_-)$, $IS_+^{L\Sigma_1}(A, G_-)$, $IS_+^{U\Delta_0}(A, G_-)$, etc. are consistent for all nice A , and that once again the same is false for $IS_+^{L\Pi_1}(A, G_-)$.

Remark 6. For simplicity, this paper has employed the definition of G_0 , introduced in Section 2. Its definition was quite arbitrary, and our main theorems trivially extend for more general definitions. For instance, say the function symbol $\hat{f}[i,j]$ (from Section 5) is **slowly growing** iff $\hat{f}[i,j](x_1, x_2, \dots, x_j) \leq i 2^j \cdot \text{Max}(x_1, x_2, \dots, x_j)$. Propositions 1-4 and 8 & 9 will hold when any set of slowly growing functions is added to G_0 , and Proposition 7 is valid when any set of non-growth functions is added to G_- .

Remark 7. Let R denote any recursively enumerable axiom system that uses the language symbols of IS and which is consistent with G_0 . Clearly, there will then exist a Δ_0 formula $\text{Deriv}(x, y)$ such that Φ is a theorem of R if and only if $\exists y \text{ Deriv}(\ulcorner \Phi \urcorner, y)$. Define A_R to be the union of G_0 with the added axiom schema of the form

$$\forall y \{ \text{Deriv}(\ulcorner \Phi \urcorner, y) \supset \Phi \} \quad (13.3)$$

The system A_R is obviously nice, and Propositions 1, 3 & 7 will consequently imply that $IS(A_R)$, $IS^{\Sigma_1}(A_R)$, and $IS_+(A_R, G_-)$ are consistent systems that prove all the Π_1 theorems of R .

14. Philosophical Speculations

This article has sought to determine how powerful, in light of Godel's Incompleteness Theorem, can self-verifying systems become? Propositions 1-4 and 7-9 have illustrated *the limited but tangible power* of self-verifying formalisms, while Propositions 5 & 6 and Solovay's proof of the inconsistency of both $IS_+(PA+)$ and $IS_+(PA+, G_s)$ have formalized several restrictions on these systems.

Propositions 1 & 7 clarify the meaning of Solovay's version of the Incompleteness Theorem by showing that the inconsistent systems $IS_+(PA+, G_s)$ and $IS_+(PA+, G_0)$ become consistent when one *either* excludes Gentzen-style cuts *or* removes the successor function from the Group-1 formalism of G_s .

Our research was motivated, in part, by the perspective that humans must (in some weak sense ?) presume their own consistency to justify the mental energy expended during cogitation. The *CFA* system of Kreisel and Takeuti [KT74] differs from our approach by being a self-verifying *second order* system. It also warrants the reader's close attention.

It is difficult to list crisp philosophical conclusions that follow from the theorems of this paper because many of our self-verifying formalisms become inconsistent when their underlying definitions are changed only slightly. A possible explanation for the sharp contrast between Proposition 5's *invalid* Π_1 local reflection principle and *the quite valid close approximations to it* in Propositions 2c, 4, 8 and 9 was given in Section 10. As noted, there are valid epistemological perspectives, where a proof of a Π_1 theorem should be viewed exclusively *as a proof*, rather than also *as a theorem*!

It should be mentioned that all our formalisms in Propositions 1-4 and 8 & 9 remain consistent in the presence of Gentzen-style cuts, in the degenerate case where the sentence being cut is Δ_0 (see *). Thus, Solovay's Theorem about the inconsistency of $IS_+(PA+)$ rests on applying the cut deduction rule to sentences with unbounded quantifiers. Also the passage containing Equation (3.7) described *types of Δ_0 cut principles* such that these cut principles are valid for introspective IS-like systems *but invalid for classical axiom systems*.

One interesting question concerns the relationship between Zermelo-Fraenkel Set Theory (ZF) and the axiom system $IS(ZF)$. While ZF set theory can not prove its own consistency [Go31], all the proofs within this paper can be carried out within ZF. This is a bit interesting because it means ZF can prove its consistency is equivalent to an alternate system $IS(ZF)$, *which is self-verifying and affirms the consistency of ZF*. (Indeed, $IS(ZF)$ confirms ZF's Π_1 validity.) This raises the philosophical question of whether *when humans think* they are using an analog of ZF for reasoning, they may be actually relying *unconsciously upon* some variant of a more introspective $IS(ZF)$ (or perhaps upon $IS^{\Sigma_1}(ZF)$, $IS^{L\Sigma_1}(ZF)$, $IS^{U\Sigma_1}(ZF)$ or $IS^{U\Delta_0}(ZF)$ etc.).

To some extent, Logic appears to be analogous to Physics, where neither the wave nor particle theory convey the full truth, and individual axiomatic frameworks can convey only partial perspectives. In a similar spirit, it is awkward to both use a self-verifying system (because of its weak structure), or to employ a non-self-verifying systems (because it does not explain how humans justify the mental energy expended during cogitation without implicitly presuming their consistency). Some hybrid perspective where one can readily shift perspectives, as in the example of Physics, is perhaps needed. In this regard, we note that the pair of systems, ZF and $IS(ZF)$, do allow one to

* One way to formally extend $IS(A)$ to include Δ_0 cutting is to simply add to $IS(A)$ an axiom $\Phi \vee \neg\Phi$ for each Δ_0 sentence Φ (and verify that all our Propositions remain valid under such a modification). However, this is essentially unnecessary because $IS(A)$, without modification, can simulate a Δ_0 cut rule with only a polynomial increase in proof length. This nice property holds because if Φ is Δ_0 then $IS(A)$'s Group-2 axioms include the sentence $\text{Prf}_A(\ulcorner \Phi \vee \neg\Phi \urcorner) \supset \Phi \vee \neg\Phi$.

readily shift perspectives in that:

- i) ZF is strong, and it can comprehend (through its own internalized version of Model Theory) that ZF is consistent if and only if IS(ZF) is consistent.
- ii) IS(ZF) is self-verifying and clearly able to employ ZF at the level of comprehending ZF's Π_1 theorems.

Thus, the pair of systems, ZF and IS(ZF), has both a strong and self-verifying component, with each component understanding quite well the significance of the other !!! (Curiously, the ZF *half* of this 2-part system can be viewed as performing deduction *with Gentzen-style cuts*, if ZF is defined to include the axiom $\Phi \vee \neg \Phi$ for each sentence Φ . On the other hand by the combination of Solovay's *negative* and Proposition 7's *positive* result, the IS(ZF) fragment of this system can not consistently employ deduction with Gentzen cuts, unless all growth functions, including Successor, are removed from IS(ZF)'s Group-1 axioms !)

The results of this paper are likely to also have some significance even if one subscribes to the philosophical position that human thought is too complicated to be encapsured by any single (or pair) of r.e. axiom systems. For example, consider an axiom system A which is constructed by taking the union of a countable class of transcending axiom systems $A_1 \subset A_2 \subset A_3 \subset \dots$ where each A_i *but not* A is r.e. There was insufficient space to explore such structures in this paper, but Propositions 1 thru 9 clearly have some implications for them, since they define respects in which the sub-components A_i can be required to be self-verifying.

Finally, it should be noted that the title of this chapter was "Philosophical Speculations" rather than "Concluding Remarks". The former title was selected because it is difficult to list *crisp philosophical conclusions* following from our theorems when many of the self-verifying formalisms become inconsistent after their underlying definitions are changed only slightly.

Thus, the most important conclusion of this paper may be merely that several provably correct self-verifying axiom systems become inconsistent when their underlying definitions are changed only slightly.

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APPENDIX A

This appendix will show that for each nice A , it is possible to encode $\text{Prf}_{\text{IS}(A)}(x,y)$ as a Δ_0 formula. For simplicity, our discussion will focus exclusively on the $\text{IS}(A)$ system from Propositions 1 & 2. An easy generalization will imply that for any $G \supseteq G_0$, it is also possible to provide the $\text{IS}_d^*(A, G)$ systems in Proposition 3 thru 9 with a Δ_0 formula for encoding $\text{Prf}_{\text{IS}_d^*(A,G)}(x,y)$.

We will often employ the following notation.

- i) $\text{UNION}(A)$ designates the union of $\text{IS}(A)$'s Group-1 and Group-2 axioms.
- ii) $\text{ExPrf}_B(x,y,z)$ designates the formula that y is a semantic tableaux proof of x using the axiom system B *plus the added axiom whose Godel number equals z* .
- iii) $\text{SUBST}_i(x,y)$ denotes Godel's classic substitution relation which yields TRUE when x is a Godel number of a formula that contains a free variable $\hat{v}[i]$, y is the Godel number of a formula generated by replacing each appearance of " $\hat{v}[i]$ " with the Godel number x , and where $\hat{v}[i]$ is always free in x and it has at least one such appearance. (It is convenient to define $\text{SUBST}_i(x,0)$ to be TRUE when x fails one of the preceding conditions, i.e. when x is not the Godel number of a formula, it does not contain an appearance of $\hat{v}[i]$, or some appearance of $\hat{v}[i]$ is not free.)

Our notation will imply that if H is the Godel number of the sentence specifying $\text{IS}(A)$'s Group-3 axiom then $\text{ExPrf}_{\text{UNION}(A)}(x,y,H)$ essentially formalizes $\text{Prf}_{\text{IS}(A)}(x,y)$. The relation SUBST will be employed to define formally the integer H , as will be shown in Equations (A.2) & (A.3).

Point of Clarification. Since it does not recognize multiplications as a function, $\text{IS}(A)$ will clearly be too weak to prove that $\forall x \exists y \text{SUBST}_i(x,y)$. However, $\text{IS}(A)$ will be able to prove for *any fixed integer $\bar{k}$* that $\exists y \text{SUBST}_i(\bar{k}, y)$. It will turn out that the latter will be sufficient for $\text{IS}(A)$ to formally define the *particular instance of the reflection principle* needed to define H .

Theorem A.1. Recall that the class of majorized formulae were defined in Lemma 5.1. This class is broad enough so that:

- a) $\text{SUBST}_i(x,y)$ can be encoded by a majorized formula;
- b) the three statements that a Godel number x is an axiom belonging to one of the conventional axiom systems of PA , PA^+ , or ZF can be encoded as majorized formulae;
- c) for any axiom system B which can be encoded by a majorized formula, the expression $\text{Prf}_B(x,y)$ can also be encoded as a majorized formula;
- d) for any nice A , the expression $\text{ExPrf}_{\text{UNION}(A)}(x,y,z)$ (and hence also $\text{Prf}_{\text{UNION}(A)}(x,y)$) can be encoded as majorized formulae.

Before proving Theorem A.1, its significance should be explained. The combination of Lemma 5.1 and Theorem A.1 imply that $\text{SUBST}_i(x, y)$, $\text{Prf}_{\text{UNION}(A)}(x, y)$ and $\text{ExPrf}_{\text{UNION}(A)}(x, y, z)$ can be encoded as Δ_0 formulae. This implies that Equation (A.1) can be encoded as a Π_1 formula.

Notation in Equation (A.1). Our notation has been informal in this article, and we usually omit Section 5's $\hat{\vee}$ symbol. For instance, the symbols y and z in Equation (A.1) should technically be written as $\hat{\vee}[y]$ and $\hat{\vee}[z]$, but we omit the tiresome notation because their implicit meaning is clear. One exception to this rule is that the formal symbol $\hat{\vee}[i]$ is employed in Equation (A.1) because otherwise it is not evident that the designated variable has the same index as the SUBST_i relation. Essentially, this is the only exception appearing in this article.

$$\forall y \rightarrow \{ \text{Prf}_{\text{UNION}(A)}(\perp, y) \vee [\exists z < y \text{SUBST}_i(\hat{\vee}[i], z) \wedge \text{ExPrf}_{\text{UNION}(A)}(\perp, y, z)] \} \quad (\text{A.1})$$

Let J denote the Godel number of Equation (A.1). This Godel number will enable us to formally define the construct $\text{Prf}_{\text{IS}(A)}(x, y)$, which was employed but not given a formal definition in Sections 2 thru 7. Equation (A.3) is the formal definition of $\text{Prf}_{\text{IS}(A)}(x, y)$, and (A.2) is $\text{IS}(A)$'s corresponding Group-3 axiom (of the generic form " $\forall y \rightarrow \text{Prf}_{\text{IS}(A)}(\perp, y)$ "). (These equations enable us to formalize $\text{Prf}_{\text{IS}(A)}(x, y)$ because the only integer z satisfying $\text{SUBST}_i(J, z)$ in equations (A.2) & (A.3) is (A.2)'s Godel number.) Since SUBST , Prf and ExPrf can be given Δ_0 definitions, Equation (A.2) is a Π_1 sentence, and (A.3) is a Δ_0 formula. These definitions of provability and of $\text{IS}(A)$'s Group-3 axiom will thus meet the requirements of the closing paragraph in Section 5.

$$\forall y \rightarrow \{ \text{Prf}_{\text{UNION}(A)}(\perp, y) \vee [\exists z < y \text{SUBST}_i(J, z) \wedge \text{ExPrf}_{\text{UNION}(A)}(\perp, y, z)] \} \quad (\text{A.2})$$

$$\text{Prf}_{\text{UNION}(A)}(x, y) \vee \{ \exists z < y \text{SUBST}_i(J, z) \wedge \text{ExPrf}_{\text{UNION}(A)}(x, y, z) \} \quad (\text{A.3})$$

Clarifying Comment 1. Equation (A.3) appears to be the simplest possible representation of $\text{Prf}_{\text{IS}(A)}(x, y)$ as a Δ_0 formula. The reason is easiest to appreciate if K denotes the unique Godel number satisfying $\text{SUBST}_i(J, K)$. Let $y < K$ encode a proof of the sentence x from the axiom system $\text{UNION}(A)$. Then y will not satisfy the curly bracket expression in Equation (A.3), and the formula $\text{Prf}_{\text{UNION}(A)}(x, y)$ in Equation (A.3) is necessary to indicate that y is a valid proof from the axiom system $\text{IS}(A)$. On the other hand, any proof y that includes $\text{IS}(A)$'s Group-3 axiom will have a Godel number greater than K . Thus, the curly bracket expression in Equation (A.3) is sufficient to test for whether y is this second type of proof of x .

Clarifying Comment 2. The preceding discussion has shown how the combination of Parts (a) & (d) of Theorem A.1 enables us to construct a Δ_0 definition for $\text{Prf}_{\text{IS}(A)}(x,y)$. It did not employ Parts (b) or (c) of Theorem A.1, and their significance should now be explained. Combined with Lemma 5.1, Part (b) implies that the axiom systems PA, PA+ and ZF have Δ_0 definitions. Thus, these axiom systems meet the requirements for Propositions 1 thru 9 to be applicable to $\text{IS}(\text{PA})$, $\text{IS}(\text{PA}+)$ and $\text{IS}(\text{ZF})$. The combination of Lemma 5.1 and Part (c) of Theorem A.1 imply that for each nice axiom system A, $\text{Prf}_A(x,y)$ can be encoded as a Δ_0 formula. This fact is important because it justifies our presumption that all $\text{IS}(A)$'s Group-2 axioms can be encoded as Π_1 sentences (i.e., it justifies the sentence which had followed Equation (2.1)).

The remainder of this Appendix will be devoted to proving Theorem A.1's four claims. As is often the case with the construction of Gödel encodings, the formal details of the proofs are quite tedious. These details are included only for the sake of completeness. The main theorems of Sections 3-14 are clearly of greater conceptual interest. It is strongly recommended that they be examined before the remainder of this Appendix. (Indeed, since Section 5 showed $\text{Prf2}(x,y,z)$ can be trivially encoded with a Δ_0 definition, readers not caring about the technical distinction between Prf and Prf2 can reasonably choose to omit the entire remainder of this Appendix.)

Recall that a byte is defined in this paper as an unit consisting of six bits. The eight atomic function of G_0 , listed in Section 2, manipulated essentially binary encodings of integers (as bit strings). Since this Appendix will be studying mostly byte strings (rather than bit strings), it will be convenient to define several further compound functions, built from these atomic functions: Eighteen such compound functions, mostly for byte manipulation, are defined below. (Since only 54 letters appeared in the alphabet defined in Section 5, we may assume without loss of generality that no alphabetic symbol is encoded by the byte of zero.)

- 1) $\text{Count}(x,b)$ = the number of repetitions of the byte b in the byte string x {formally encoded as $\text{Count}(x,b) = \text{StringCount}(x,b,6)$ }.
- 2) $\text{Log}^+(x)$ designating $\lceil \log_2(x+1) \rceil$. {formally encoded as $\text{Log}^+(x) = \text{StringCount}(x,0,1) + \text{StringCount}(x,1,1)$ }.
- 3) $\text{Length}(x)$ designating x 's byte length {formally encoded as $\text{Length}(x) = (\text{Log}^+(x)+5)/6$ }
- 4) $\text{Scalar}_k(x)$: comprising a family of functions that for *any fixed constant* k represents the scalar product of $k \cdot x$. {Unlike normal multiplication, scalar multiplication is a function under $\text{IS}(A)$ because it is defined by $\text{Scalar}_0(x) = 0$ and $\text{Scalar}_{k+1}(x) = \text{Scalar}_k(x) + x$. The

notation $2x, 3x, 4x, \dots$ will be an abbreviation in this paper for $\text{Scalar}_2(x), \text{Scalar}_3(x), \text{Scalar}_4(x), \dots$ }

- 5) Substring (x, i, j) designating the substring of x consisting of its i -th through j -th bytes put into right-justified form {formally encoded as $\text{Shift}(\text{Extract}(x, 6i-5, 6j), 6i-6)$ }
- 6) Byte (x, i) designating x 's i -th byte {formally encoded as $\text{Substring}(x, i, i)$ }.
- 7) Bit (x, i) designating x 's i -th bit {formally encoded as $\text{Shift}(\text{Extract}(x, i, i), i-1)$ }.
- 8) Power $(x) =$ the greatest number $y \leq x$ which is a power of 2 {constructed by setting $\text{Power}(x) = \text{Extract}(x, \text{Log}^+(x), \text{Log}^+(x))$ }
- 9) Ones $(x) = 2 \cdot \text{Power}(x) - 1$ {i.e. a sequence of "1" bits whose bitlength equals $\text{Log}^+(x)$ }.
- 10) Maximum (x, y) which can be defined as $x + (y - x)$ {since Section 2's definition of subtraction indicated that $y - x = 0$ when $x \geq y$ }.
- 11) Minimum (x, y) defined similarly as $x + y - \text{Maximum}(x, y)$.
- 12) And $(x, y) =$ bitwise AND of x and y {constructed by setting $\text{And}(x, y) = \text{Andreverse}(x, y, 1, 1)$ }.
- 13) Indicate $(x, y) = x$ when $y \neq 0$, and it equals zero when $y = 0$. {formally encoded as $\text{Indicate}(x, y) = x / (y / y)$ since section 2's definition of division indicated that $a/0 = 0$.) }
- 14) Find (x, i, b) designating the address in the byte string x of the i -th occurrence of the byte b (if such an occurrence exists), and $\text{Find}(x, i, b) = 0$ otherwise. {Let K be any integer greater than 64 (so that it is impossible for K to designate a byte). Then $\text{Find}(x, i, b)$ can be encoded as $\text{Address}(x, b, K, i, 6) / 6$. }
- 15) Next $(x, i, b) =$ the address in the byte string x of the first occurrence of the byte b to the right of the address i (if such an occurrence exists), and $\text{Next}(x, i, b) = \text{Length}(x)$ otherwise. Since Section 2 defined $u - v$ as equal to zero when $u \leq v$, $\text{Next}(x, i, b)$ can be encoded as $\text{Find}[x, \text{Count}(\text{Substring}(x, 1, i), b) + 1, b] + \text{Indicate}[\text{Length}(x), 1 - \text{Find}(x, \text{Count}(\text{Substring}(x, 1, i), b) + 1, b)]$.
- 16) Bracket $(x, i) =$ the address in the byte string x of the first occurrence of either the byte "{" or "}" to the right of the address i . It is formally encoded as $\text{Minimum}[\text{Next}(x, i, "{"), \text{Next}(x, i, "}")]$.
- 17) ReverseOrder (x) designating the bit-string s of length $L = \text{Log}^+(x)$ such that $\text{Bit}(s, i) = \text{Bit}(x, L+1-i)$ for each $i < L+1$. ReverseOrder (x) is formally encoded as $\text{Andreverse}(x, \text{Ones}(x), 1, \text{Log}^+(x))$.

- 18) $\text{ReverseAddress}(x, b, c, j)$ indicating the greatest integer k where $\text{Substring}(x, k, \text{Length}(x))$ contains j more appearances of the byte b than of the byte c (if such k exists), and $\text{ReverseAddress}(x, b, c, j) = 0$ if k does not exist {trivially encoded with the functions Address and ReverseOrder , since ReverseAddress is essentially the same as the function Address with the bytes processed in reverse order}.

Recall that Section 5 indicated that a tree, whose root contains the string P and where the subtrees $\{\alpha_1\}, \{\alpha_2\}, \{\alpha_3\}, \dots \{\alpha_n\}$ are the children of the root, will be encoded as the parenthesized expression $\{ P \{ \alpha_1 \} \{ \alpha_2 \} \{ \alpha_3 \} \dots \{ \alpha_n \} \}$. The symbol P will henceforth be called the root's *principle string*, and it will be assumed to be a non-empty string containing no instances of the special punctuating bytes "{" or "}". Each node $\{\alpha_i\}$ descending from the root will have an analogous recursively defined construction.

Lemma A.2. There exists a Δ_0 formula $\text{TreeCheck}(s)$ which will determine whether an arbitrary byte string s is a parenthesis tree (as defined above).

Proof. It is easy to verify that s is a parenthesized tree if and only if it satisfies the following five conditions:

- i) the first byte in s is "{" and its last byte is "}";
- ii) Every "}" byte, except for the last byte of s , is immediately followed by a byte which is either "{" or "}" ;
- iii) No "{" is immediately followed by the byte "}" or "{" (since at the very least a nonempty principle string will separate these two bytes);
- iv) the same number of "{" and "}" bytes appear in s ;
- v) For each $i < \text{Length}(s)$, there are more appearances of "{" than "}" among the first i bytes of s .

The formal encoding of these five conditions as Δ_0 formulae are indicated below:

- i) $\text{Byte}(s, 1) = \{ \wedge \text{Byte}(s, \text{Length}(s)) = \}$
- ii) $\forall i < \text{Length}(s) : \text{Byte}(s, i) = \} \supset \text{Byte}(s, i+1) = \{ \text{ or } \}$
- iii) $\forall i < \text{Length}(s) : \text{Byte}(s, i) = \{ \supset \text{Byte}(s, i+1) \neq \} \text{ or } \{$
- iv) $\text{Count}(s, \{) = \text{Count}(s, \{)$
- v) $\forall i < \text{Length}(s) : i=0 \vee \text{Count}(\text{Substring}(s, 1, i), \{) > \text{Count}(\text{Substring}(s, 1, i), \}$